

Experiment 3.4

Determination and plotting of Force between two parallel current carrying conductors carrying equal currents

Code File Name: Exp3_4_ParallelConductors_EqualCurrents.m

Code

```
clc;
mu0 = 4*pi*1e-7;
I1 = 3;
I2 = 3;
d = linspace(0.1,2,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between conductors (equal currents)');
```

Output Graph

